

Notes: Biasi & Sarsons (QJE, 2022)

Flexible Wages, Bargaining, and the Gender Gap

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1 Big picture

- **Question.** Does the introduction of flexible pay and individual bargaining widen the gender wage gap?
- **Setting.** Wisconsin public school teachers before and after Act 10, a 2011 reform that removed unions' authority to bargain over teacher salary schedules after preexisting collective-bargaining agreements expired.
- **Research design.** The paper exploits quasi-exogenous variation in when school districts became exposed to flexible pay, driven by preexisting CBA expiration and extension dates.
- **Main result.** When flexible pay begins, women earn less than comparable men with the same experience, credentials, and teaching assignment. The aggregate post-flexibility gap is about 0.8 percent and grows with time since CBA expiration.
- **Mechanism evidence.** Survey evidence suggests women bargain less often over salary, especially when the superintendent is male. The wage gap is smaller or absent under female principals/superintendents.
- **Alternative explanations.** The gap is unlikely to be explained by value added, observable mobility, composition changes, or teacher exit patterns. Men receive returns to value added after flexible pay; women do not.
- **Economic takeaway.** Wage flexibility can create room for gender inequality even when rigid salary schedules previously equalized pay conditional on credentials and seniority.

2 Incremental contribution of the paper

1. **Field evidence on bargaining and pay flexibility.** Laboratory studies show women negotiate less often, but field evidence is difficult because workers sort into jobs. This paper uses an institutional shock that changes bargaining requirements for an existing workforce.
2. **Institutional variation with staggered timing.** The paper turns Wisconsin Act 10 into a staggered-difference design using CBA expiration and extension dates, rather than comparing all districts before/after the law at the same time.
3. **Separates flexible pay from worker sorting.** The same occupation, public school teachers, moves from rigid union schedules to flexible pay, limiting confounding from broad occupational differences.
4. **Links pay outcomes to negotiation behavior.** The administrative wage results are paired with a teacher survey on bargaining attempts, future negotiation intentions, perceived comfort, and perceived success.
5. **Studies institutional mediators.** The paper shows that leadership gender matters: gaps are smaller under female principals and superintendents, suggesting that counterpart identity and organizational context shape bargaining outcomes.
6. **Tests alternative productivity and mobility channels.** The paper uses teacher value added, job mobility, commuting-zone variation, and survey measures of outside options to show that bargaining rather than productivity or mobility is the more plausible channel.

3 Data and measurement

3.1 Teacher administrative data

The administrative data contain Wisconsin public school teachers from 2007 to 2016. The key variables are teacher salary, gender, experience, age, degree credentials, full-time-equivalent status, school district, grade level, subject assignment, school, principal, superintendent, and teacher value added for teachers in tested grades and subjects.

Interpretation: The setting is especially useful because pre-Act-10 pay schedules made salary largely a function of observable credentials. After the reform, districts could pay teachers different amounts even within the same observable credential cells.

3.2 Policy shock: Wisconsin Act 10

Before Act 10, unions bargained salary schedules with school districts. After Act 10, unions lost authority over salary schedules once the existing CBA expired. Districts could then set pay flexibly, including individual adjustments outside the old schedule.

The timing of exposure is district-specific:

$$\text{FlexiblePay}_{d,t} = 1\{t \geq \text{CBA expiration/extension date}_d\}.$$

Interpretation: The expiration date is the key treatment timing. The paper uses both original CBA expirations and extensions, and checks robustness using original expirations as instruments for extensions.

3.3 Outcome and core wage gap

The main outcome is the natural logarithm of teacher salary, multiplied by 100 and measured in full-time-equivalent units:

$$y_{i,d,t} = 100 \times \log(\text{FTE-adjusted salary}_{i,d,t}).$$

The main coefficient of interest is the differential post-expiration change for female teachers:

$$\text{Female}_i \times \text{Post}_{d,t}.$$

Interpretation: Because the outcome is in log points times 100, coefficients are approximately percentage-point wage differences.

3.4 Survey data

The authors surveyed Wisconsin public school teachers to measure bargaining behavior. The survey asks whether teachers negotiated salary at the start of the relationship, negotiated after the start, expect to negotiate salary/classroom assignment/nonteaching duties, and why they did not negotiate.

Interpretation: The survey cannot be linked to administrative wage records, but it provides direct mechanism evidence on whether women bargain less often and whether leadership gender changes the bargaining environment.

4 Empirical specifications

4.1 Event-study salaries by gender

The first descriptive specification estimates salary paths by gender relative to the expiration/extension date:

$$y_{i,d,t} = \alpha_i + \lambda_t + \sum_{s \neq -1} \delta_s^m \mathbf{1}\{t - T_d = s\} \cdot \text{Male}_i + \sum_{s \neq -1} \delta_s^f \mathbf{1}\{t - T_d = s\} \cdot \text{Female}_i + X'_{i,t} \beta + \varepsilon_{i,d,t}.$$

Interpretation: This event-study shows whether male and female salaries move similarly before flexible pay and diverge afterward.

4.2 Gender gap after CBA expiration

The main reduced-form model can be summarized as:

$$y_{i,d,t} = \alpha_d + \lambda_t + \theta Female_i + \gamma Post_{d,t} + \delta(Female_i \times Post_{d,t}) + X'_{i,t}\beta + \varepsilon_{i,d,t}.$$

The coefficient δ captures the post-flexible-pay change in the female-male salary gap.

Interpretation: A negative δ means women lose salary relative to men after their district receives pay flexibility.

4.3 Dynamic post effects

The dynamic version replaces a single post indicator with separate indicators for one, two, three, four, and five years after expiration:

$$y_{i,d,t} = \dots + \sum_{k=1}^5 \delta_k(Female_i \times \mathbf{1}\{k \text{ years post}\}) + \varepsilon_{i,d,t}.$$

Interpretation: The dynamic model asks whether the gender gap emerges immediately or accumulates over time. The largest estimates appear several years after flexibility begins.

4.4 Mechanism and heterogeneity

The paper estimates variants of the main model separately by:

- district type: no schedule versus schedule and gender-specific schedules;
- leadership gender: male/female principal and superintendent;
- teacher age and seniority;
- value added;
- mobility status and commuting-zone outside options.

Interpretation: These heterogeneity tests distinguish bargaining from alternative mechanisms such as productivity, ability, and mobility.

5 Identification and empirical design

5.1 Source of identifying variation

Identification comes from staggered variation in CBA expiration or extension dates across districts. Districts become exposed to flexible pay at different dates because their pre-Act-10 contracts ended at different times.

Interpretation: The design compares changes in the gender wage gap before and after flexible-pay exposure, using later-treated districts as controls for earlier-treated districts.

5.2 Parallel-trends logic

The key assumption is that, absent the introduction of flexible pay, male and female salaries would have continued on parallel paths across districts with different expiration dates. The event-study figures test this by showing little pre-trend in the gender gap before expiration.

Interpretation: The identifying variation is credible because the contracts were negotiated before the reform and salary gaps were not trending differentially before treatment.

5.3 Extensions and 2SLS

Some districts extended their CBAs. Extensions may be more endogenous than original expirations. The paper therefore estimates:

- OLS using actual expiration/extension dates;
- OLS using original expiration dates only;
- 2SLS using original expiration timing as an instrument for extension timing.

Interpretation: Similar estimates across these approaches support the claim that extension choices are not driving the main result.

5.4 Mechanism validity

The survey evidence supports bargaining as a mechanism: women are less likely to negotiate salary and more likely to report discomfort negotiating. The wage gap is concentrated when leaders are men and disappears or attenuates when leaders are women.

Interpretation: The survey does not prove that bargaining explains the entire wage gap, but it provides direct behavioral evidence consistent with the administrative wage results.

5.5 Threats and responses

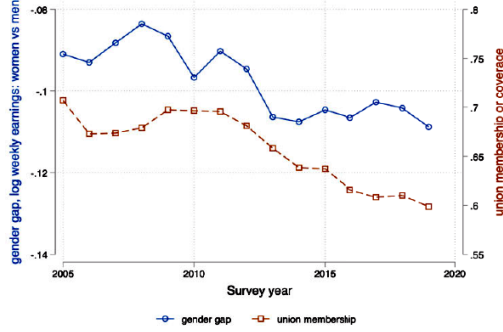
- **Productivity changes.** Controlled using teacher value added; women do not have lower value added, and the wage gap remains.
- **Mobility differences.** Mobility regressions and Figure VI suggest observed mobility is not the main explanation.
- **Composition changes.** Tests of teacher attrition and characteristics show limited evidence that sample composition drives the gap.
- **District selection into schedules.** Table III separates schedule and nonschedule districts and shows that no-schedule districts generate a larger wage gap.
- **Leadership selection.** Principal and superintendent gender are not randomly assigned, so leadership heterogeneity is interpreted as mechanism evidence rather than causal proof by itself.

6 Figures and tables

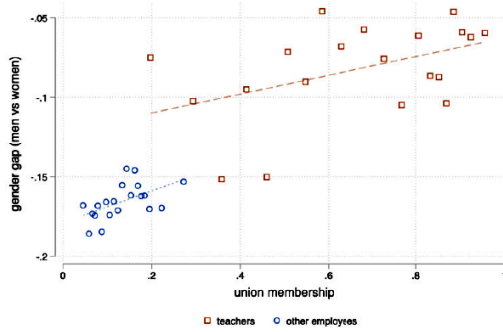
6.1 Figure I: Gender Wage Gap and Unionization; Table I: Mean Observable Characteristics, Male and Female Teachers

Read: Figure I documents that higher union membership is associated with smaller gender wage gaps, and Table I shows that male and female teachers are observably different in some dimensions but broadly comparable within the administrative data.

Economic message: Figure I motivates the institutional channel: weakening collective bargaining may widen the wage gap. Table I shows why the empirical analysis controls for experience, credentials, grade, and subject assignment.



(A) The gender wage gap and union membership over time (public school teachers)



(B) Gender wage gaps and union membership (public school teachers and other employees)

FIGURE I

Figure I: Gender Wage Gap and Unionization. Unionization and gender wage gaps move together in ways consistent with unions compressing gender pay gaps.

	2007–2011			2012–2016		
	Males	Females	Diff.	Males	Females	Diff.
Experience (yrs)	15.0	14.5	0.51*** (0.042)	14.2	13.9	0.28*** (0.039)
Age	42.9	43.3	-0.38*** (0.047)	42.4	42.5	-0.053 (0.045)
Value added	-0.0023	-0.00023	-0.0021*** (0.00069)	-0.0012	-0.000067	-0.0011** (0.00048)
Salary (\$1,000)	51.3	51.2	0.063 (0.051)	54.2	53.9	0.33*** (0.053)
Full-time equivalents	98.0	97.1	0.90*** (0.057)	97.1	97.0	0.096* (0.054)
Full-time	0.94	0.92	0.015*** (0.0011)	0.92	0.92	-0.0036*** (0.0012)
Ever moves	0.11	0.10	0.013*** (0.0013)	0.16	0.14	0.016*** (0.0015)
Leaves sample	0.066	0.064	0.0028*** (0.0011)	0.074	0.073	0.00073 (0.0013)
BA	0.49	0.48	0.0036* (0.0022)	0.47	0.46	0.0072*** (0.0021)
Master	0.50	0.51	-0.0049** (0.0022)	0.52	0.53	-0.0098*** (0.0021)
PhD	0.0022	0.00095	0.0012*** (0.00016)	0.0039	0.0014	0.0025*** (0.00020)
Grade level						
Elementary	0.21	0.49	-0.28*** (0.0021)	0.22	0.49	-0.27*** (0.0021)
Middle	0.30	0.25	0.043*** (0.0019)	0.29	0.25	0.046*** (0.0019)
High	0.56	0.25	0.31*** (0.0020)	0.54	0.24	0.30*** (0.0019)
Subject						
Math	0.12	0.057	0.061*** (0.0011)	0.12	0.062	0.062*** (0.0011)
Science	0.79	0.82	-0.038*** (0.0017)	0.78	0.83	-0.046*** (0.0017)
Reading	0.011	0.041	-0.029*** (0.00077)	0.011	0.041	-0.030*** (0.00077)
Observations	267,962			279,443		

Notes. The table shows mean characteristics of male and female teachers, and the differences in means (standard errors in parentheses) for the years 2007–2011 (columns (1)–(3)) and 2012–2016 (columns (4)–(6)). *Salary* is a teacher's yearly salary. *Value added* is a measure of teacher quality, described in Section III A. *Ever moves* is an indicator equal to 1 if a teacher has moved to a new school from $t-1$ to t . *Leaves sample* is an indicator equal to 1 if a teacher no longer appears in the sample in $t+1$. Elementary, Middle, and High

Table I: Mean Observable Characteristics. Observable teacher characteristics differ by gender, so controls are central to interpreting the wage-gap results.

6.2 Figure II: Share of Teachers, by Expiration and Extension Dates of CBAs; Table II: Gender Pay Gap after a CBA Expiration/Extension: OLS and 2SLS

Read: Figure II shows the distribution of treatment timing. Table II gives the core estimates: the female salary penalty emerges after CBA expiration/extension and grows in later post years.

Economic message: The figure validates that treatment timing is staggered; the table quantifies the main treatment effect.

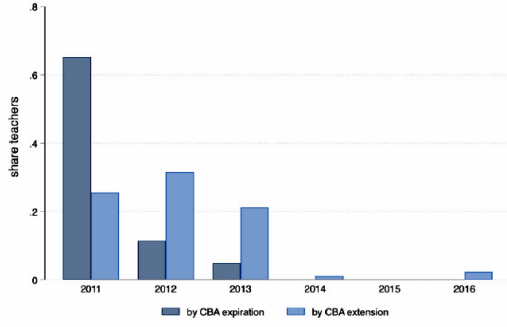


FIGURE II

Share of Teachers, by Expiration and Extension Dates of CBAs

The figure shows the share of teachers covered by collective bargaining agreements (CBAs) with staggered expiration and extension dates. Figure II: CBA expiration and extension timing. Staggered exposure to flexible pay creates the design.

	Extensions		Expirations		2SLS, extensions	
	(1)	(2)	(3)	(4)	(5)	(6)
Female	-0.121 (0.120)	-0.121 (0.120)	-0.097 (0.116)	-0.097 (0.116)	-0.097 (0.116)	-0.097 (0.116)
Female × Post Extension	-0.286*** (0.094)				-0.348*** (0.111)	
Female × 1 year(s) post		-0.062 (0.117)		-0.014 (0.114)		-0.071 (0.295)
Female × 2 year(s) post		-0.285* (0.148)		-0.269** (0.136)		-0.672* (0.358)
Female × 3 year(s) post		-0.073 (0.156)		-0.248* (0.143)		-0.015 (0.328)
Female × 4 year(s) post		-0.705*** (0.191)		-0.347* (0.186)		-0.566 (0.566)
Female × 5 year(s) post		-0.718*** (0.228)		-0.600*** (0.179)		-0.843 (0.736)
Female × Post Expiration			-0.281*** (0.089)			
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Yr. × Exp yr.	Yes	Yes	Yes	Yes	Yes	Yes
N	444,111	444,111	444,111	444,111	444,111	444,111
# districts	247	247	247	247	247	247

Notes. The dependent variable is the natural logarithm of salary per year, in full-time equivalency units and multiplied by 100. The variable Female equals 1 for female workers, the variable Post Expiration equals 1 for years following the expiration of a CBA, and the variable Post Extension equals 1 for years following the expiration of a CBA or its extension. The variables X Year(s) × Post equal 1 for observations X years after an extension (in columns (2) and (6)) or an expiration (column (4)). Columns (1)-(4) estimate OLS, columns (5) and (6) estimate 2SLS, with Post Expiration as an instrument for Post Extension. Controls include fixed effects for the district, number of years of experience, highest education degree, grade level (elementary, middle, high), and subject (math, reading, and others), alone and interacted with an indicator for years after the expiration of a CBA. All specifications also include year fixed effects interacted with CBA expiration and extension year effects. Standard errors in parentheses are clustered at the district level. * $p \leq .1$, ** $p \leq .05$, *** $p \leq .01$.

Table II: OLS and 2SLS estimates. Flexible pay lowers women's salaries relative to comparable men.

6.3 Figure III: Raw Salaries of Men and Women; Figure IV: Flexible Pay and the Gender Wage Gap

Read: Figure III plots raw salary paths around expiration dates. Figure IV plots conditional event-study estimates by gender and the gender-gap event study.

Economic message: The two figures show that men and women are on similar pretrends, then diverge after flexible pay begins.

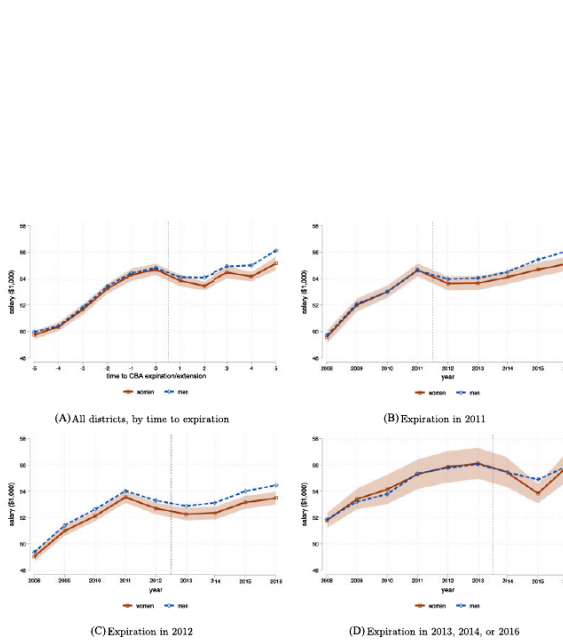
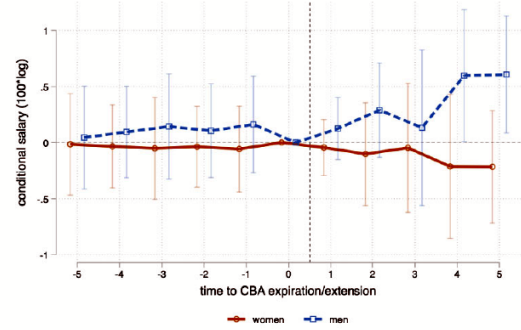


FIGURE III

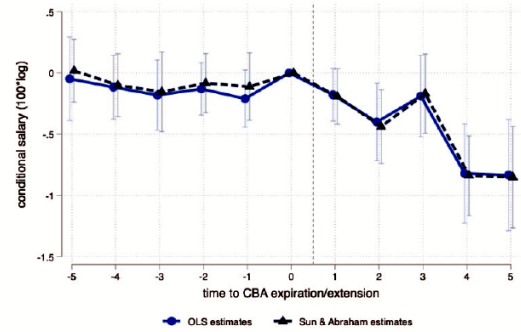
Raw Salaries of Men and Women

Panel A shows the unconditional salaries of male and female teachers relative to the year a CBA or its extension expired (denoted by $t = 0$). Panels B–D show unconditional salaries of male and female teachers, by year and separately by time of expiration/extension of districts' CBAs. Shaded areas represent confidence intervals for the female-male difference in salaries.

Figure III: Raw salaries. Raw salaries diverge after CBA expiration.



(A) Salaries of men and women, by time to expiration/extension of CBAs



(B) Gender gap in salaries, by time to CBA expiration/extension

FIGURE IV

Figure IV: Flexible pay and the gap. Conditional event studies show a growing female penalty after flexible pay.

6.4 Table III: Gender Pay Gap after CBA Expiration/Extension, by District Type; Table IV: Gender Pay Gap after CBA Expiration/Extension, by Principal and Superintendent Gender

Read: Table III shows the gap is larger in no-schedule districts than in schedule districts. Table IV shows the gap is concentrated under male principals and superintendents.

Economic message: These tables explain where flexibility matters most: less schedule rigidity and male bargaining counterparts are associated with larger gender gaps.

TABLE III
GENDER PAY GAP AFTER A CBA EXPIRATION/EXTENSION, BY DISTRICT TYPE

	Baseline			Gender-specific schedule		
	No schedule (1)	Schedule (2)	Diff. (3)	No schedule (4)	Schedule (5)	Diff. (6)
Female	-0.156 (0.162)	-0.100 (0.178)	-0.103 (0.178)	0.715 (0.327)	-0.146 (0.337)	0.144 (0.324)
Female $\times$ Post Extension	-0.318** (0.139)	-0.299** (0.144)	-0.288** (0.143)	-0.820* (0.483)	0.680 (0.582)	0.290 (0.628)
Female $\times$ No sched			-0.055 (0.239)		0.108 (0.249)	
Female $\times$ No sched $\times$ Post			-0.029 (0.199)		-0.746 (0.741)	
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Yr. $\times$ Exp yr.	Yes	Yes	Yes	Yes	Yes	Yes
Exper. $\times$ Female $\times$ Post Ext	No	No	No	Yes	Yes	Yes
N	176,917	220,414	397,439	176,917	220,414	397,331
# districts	81	99	180	81	99	180

Notes. The dependent variable is the natural logarithm of salary per year, in full-time equivalency units and multiplied by 100. The variable *Female* equals 1 for female workers and the variable *Post Extension* equals 1 for years following the expiration of a CBA or its extension. Columns (1) and (4) are estimated on teachers in districts that do not use a salary schedule after Act 19, and columns (2) and (5) are estimated on teachers in districts that keep a salary schedule. Columns (4)-(6) allow for the gender-specific returns to experience and education, by including fixed effects for years of experience and highest education degree, interacted with *Female* and an indicator for years after a CBA expiration/extension. Controls include fixed effects for the district, number of years of experience, highest education degree, grade level (elementary, middle, high), and subject (math, reading, and others), alone and interacted with an indicator for years after the extension of a CBA. All specifications also include year fixed effects interacted with CBA expiration and extension year effects. All columns present OLS estimates. Standard errors in parentheses are clustered at the district level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table III: District type. No-schedule districts have the strongest gender-gap response.

TABLE IV
GENDER PAY GAP AFTER A CBA EXPIRATION/EXTENSION, BY PRINCIPAL AND SUPERINTENDENT GENDER

	Principal			Superintendent		
	Male (1)	Female (2)	(3)	Male (4)	Female (5)	(6)
Female	-0.091 (0.114)	-0.200 (0.182)	-0.195 (0.176)	-0.037 (0.132)	-0.386** (0.186)	-0.450** (0.184)
Female $\times$ Post Extension	-0.412*** (0.103)	0.018 (0.167)	0.014 (0.162)	-0.453*** (0.151)	0.134 (0.243)	0.246 (0.260)
Female $\times$ Male princ			0.095 (0.175)			
Female $\times$ Male princ $\times$ Post			-0.440** (0.187)			
Female $\times$ Male super						0.428* (0.224)
Female $\times$ Male super $\times$ Post						-0.736** (0.365)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Yr. $\times$ Exp yr.	Yes	Yes	Yes	Yes	Yes	Yes
N	261,528	173,703	435,234	322,513	115,796	438,312
# districts	244	232	247	239	107	247

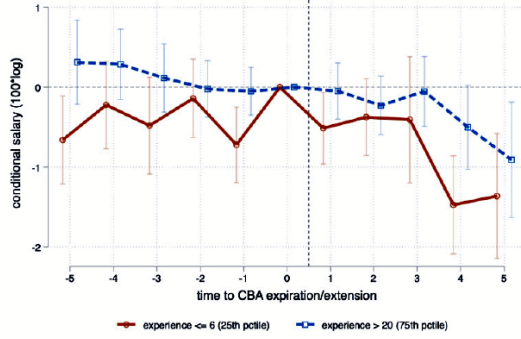
Notes. The dependent variable is the natural logarithm of salary per year, in full-time equivalency units and multiplied by 100. The variable *Female* equals 1 for female workers and the variable *Post Extension* equals 1 for years following the expiration of a CBA or its extension. The variables *Male princ* and *Male super* equal 1 for teachers in schools with a male principal and in districts with a male superintendent, respectively, in any given year. Controls include fixed effects for the district, number of years of experience, highest education degree, grade level (elementary, middle, high), and subject (math, reading, and others), alone and interacted with an indicator for years after the extension of a CBA. All specifications also include year fixed effects interacted with CBA expiration and extension year effects. All columns present OLS estimates. Columns (1) is estimated on teachers in schools with a male principal, column (2) on teachers in schools with a female principal, column (4) on teachers in districts with a male superintendent, and column (5) on teachers in districts with a female superintendent. Standard errors in parentheses are clustered at the district level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table IV: Leadership gender. The gap is larger under male principals and superintendents.

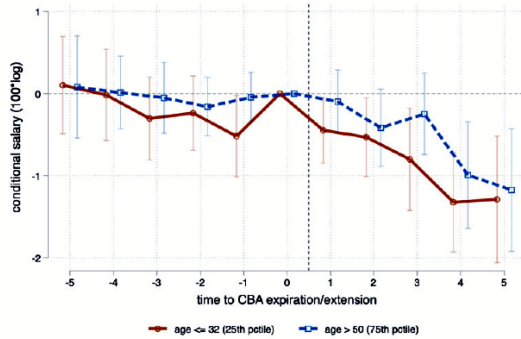
6.5 Figure V: Gender Pay Gap, by Seniority and Age; Table VII: Gender Pay Gap and Teacher Value Added

Read: Figure V shows that younger and less senior teachers experience larger gaps. Table VII shows that value added does not explain the gender gap, and that returns to value added after flexible pay are stronger for men.

Economic message: The gender gap is largest for teachers with more to gain from bargaining early in career and is not driven by lower female teacher quality.



(A) By Seniority



(B) By Age

FIGURE V

Figure V: Seniority and age. The gap is larger for younger/less senior teachers.

TABLE VII
GENDER PAY GAP AND TEACHER VALUE ADDED

	Without VA	With VA		
	(1)	(2)	(3)	(4)
Female	-0.225* (0.131)	0.281 (0.188)	0.281 (0.188)	0.283 (0.188)
Female × Post Extension	-0.326*** (0.117)	-0.062 (0.235)	-0.066 (0.235)	-0.070 (0.235)
VA (std. dev.)			0.016 (0.060)	-0.100 (0.115)
VA × Post Extension			0.131 (0.118)	
Female × VA				0.145 (0.119)
Male × VA × Post Extension				0.556** (0.240)
Female × VA × Post Extension				0.027 (0.123)
Controls	Yes	Yes	Yes	Yes
Yr × Exp yr.	Yes	Yes	Yes	Yes
N	341,608	102,499	102,499	102,499
# districts	247	247	247	247

Notes. The dependent variable is the natural logarithm of salary per year, in full-time equivalency units and multiplied by 100. The variables *Female* and *Male* equal 1 for female and male workers, respectively, and the variable *Post Extension* equals 1 for years following the expiration of a CBA or its extension. *VA* is teacher value added, described in Section III.A and standardized to have mean zero and variance one. Controls include fixed effects for the district, number of years of experience, highest education degree, grade level (elementary, middle, high), and subject (math, reading, and others), alone and interacted with an indicator for years after the extension of a CBA. All specifications also include year fixed effects interacted with CBA expiration and extension year effects. All columns present OLS estimates. Column (1) is estimated on the subsample of teachers without a VA estimate; columns (2)–(4) are estimated on the subsample of teachers with VA estimates, where we remove teachers with VA in the top and bottom 1% of the distribution. Standard errors in parentheses are clustered at the district level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table VII: Teacher value added. Productivity differences do not explain the wage gap.

6.6 Table V: Survey Answers: Means, Women versus Men, and Differences in Means

Read: Table V summarizes survey responses. Women report lower negotiation incidence and lower comfort negotiating salary, while differences in beliefs about teaching quality or future performance are less central.

Economic message: The survey supports the bargaining mechanism by showing gender differences in negotiation behavior and comfort.

TABLE V SURVEY ANSWERS: MEANS, WOMEN VERSUS MEN, AND DIFFERENCES IN MEANS				
	Women (1)	Men (2)	Difference (3)	Std. error (4)
<i>Have you ever negotiated...</i>				
w/prev employer	0.295	0.379	-0.084**	(0.019)
w/current employer, at start	0.223	0.306	-0.083**	(0.018)
w/current employer, after start	0.205	0.245	-0.040**	(0.017)
<i>Have you ever negotiated, male superintendent</i>				
w/current employer, at start	0.223	0.315	-0.093**	(0.021)
w/current employer, after start	0.204	0.273	-0.069**	(0.020)
<i>Have you ever negotiated, female superintendent</i>				
w/current employer, at start	0.216	0.285	-0.069**	(0.032)
w/current employer, after start	0.200	0.191	0.008	(0.031)
<i>If yes, was the negotiation successful?</i>				
w/prev employer	0.819	0.904	-0.085**	(0.025)
w/current employer, at start	0.709	0.814	-0.105**	(0.034)
w/current employer, after start	0.455	0.572	-0.117**	(0.042)
<i>Why did you not negotiate? (current employer, at start)</i>				
It was not possible	0.419	0.451	-0.032	(0.020)
I was not comfortable doing so	0.233	0.128	0.105**	(0.016)
It was useless	0.084	0.063	0.022**	(0.011)
I feared backlash	0.065	0.055	0.011	(0.010)
I was satisfied w/pay	0.186	0.149	0.036**	(0.015)
<i>Average likelihood that you will negotiate...</i>				
salary	3.365	3.889	-0.524**	(0.121)
classroom assignment	4.732	4.539	0.213	(0.120)
nonteaching duties	4.347	4.579	-0.232*	(0.124)

Table V, part 1: Survey means. Women are less likely to have negotiated, especially salary.

TABLE V CONTINUED				
	Women (1)	Men (2)	Difference (3)	Std. error (4)
<i>Average likelihood that you will negotiate, male superintendent</i>				
salary	3.233	3.996	-0.764***	(0.143)
classroom assignment	4.652	4.449	0.202	(0.157)
nonteaching duties	4.215	4.509	-0.293**	(0.148)
<i>Average likelihood that you will negotiate, female superintendent</i>				
salary	3.556	3.667	-0.110	(0.229)
classroom assignment	4.922	4.714	0.209	(0.237)
nonteaching duties	4.581	4.724	-0.143	(0.231)
<i>Share agreeing w/ statements</i>				
I worked in other industries	0.476	0.503	-0.027	(0.020)
I know someone who negotiated their pay	0.505	0.590	-0.085**	(0.020)
I know my colleagues' pay	0.275	0.387	-0.111**	(0.019)
I am confident talking to people I don't know	0.728	0.839	-0.110**	(0.017)
I can read subtle signals	0.890	0.884	0.006	(0.013)
I can read people's feelings	0.871	0.861	0.010	(0.014)
I have good people skills	0.888	0.883	0.006	(0.013)
My performance is above the mean	0.321	0.364	-0.044**	(0.019)
N (teachers)	2,190	843		

Notes. Mean survey answers by genders and differences between men and women, along with standard errors (in parentheses). The options under *Why did you not negotiate* are mutually exclusive choices offered to the respondents. Online Appendix D contains the full list of questions. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table V, part 2: Survey means continued. Women report less comfort and lower negotiation propensity.

6.7 Table VI: Survey Answers: Negotiating Behavior

Read: Table VI estimates gender differences in bargaining behavior conditional on controls. Women are less likely to negotiate at the start of a relationship, less likely to negotiate afterward, and less likely to expect future salary negotiation; they are more likely to report discomfort.

Economic message: The table connects the administrative gender wage gap to an observed behavioral channel: bargaining.

	Current employer				Likelihood future negotiations			
	At start (1)	At start (2)	After (3)	After (4)	Salary (5)	Salary (6)	Classroom assign. (7)	Nonteaching duties (8)
Panel A: Past and future negotiations								
Female	-0.071*** (0.022)	-0.083*** (0.025)	-0.028 (0.018)	-0.057*** (0.020)	-0.475*** (0.162)	-0.718*** (0.151)	0.273* (0.139)	-0.135 (0.133)
Female × F super		0.038 (0.046)		0.086** (0.035)		0.746** (0.354)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,836	2,784	2,836	2,784	2,836	2,784	2,836	2,836
Y mean, males	0.306		0.245		3.889		4.539	

	Successful negotiation			Reasons for not negotiating			
	Cond (1)	Uncond (2)	Cond (3)	Not possible (4)	Not comfortable (5)	Useless (6)	Fear backlash (7)
Panel B: Negotiated successfully or reasons for not negotiating							
Female	-0.132** (0.052)	-0.082** (0.020)	-0.087 (0.057)	-0.023 (0.028)	0.065** (0.029)	0.024 (0.025)	0.005 (0.019)
Female × F super			-0.134 (0.119)				
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	700	2,836	682	2,222	2,222	2,222	2,222
Y mean, males	0.904	0.249	0.814	0.565	0.210	0.215	0.131

Notes: In Panel A, the dependent variable equals 1 if a teacher negotiated their salary with the current employer at the start of the work relationship (columns (1)-(2)), or after the start (columns (3)-(4)), or the likelihood, measured as a number from 0 to 1, that the teacher will negotiate either salaries (columns (5)-(6)), classroom assignment (column (7)), or nonteaching duties (column (8)) in the future. In Panel B the dependent variable equals 1 if a teacher believed the negotiation with her current employer (at the start of the work relationship or after the start) was successful, conditional on negotiating (columns (1) and (2)) or unconditionally (column (3)) or if a teacher gives the corresponding reason as a motive for not negotiating conditional on not doing so, columns (4)-(8). Female is an indicator for female teachers. F super equals 1 for teachers in districts run by female superintendents. All specifications include controls for age group, self-reported job performance (above/below average), measures of people skills, an indicator for whether the respondent knows someone who negotiated salary, an indicator for whether the respondent knows higher colleagues' salaries, and district fixed effects. Standard errors in parentheses are clustered at the district level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table VI, part 1: Negotiation behavior. Women negotiate salary less often.

Table VI, part 2: Reasons and success. Women more often report discomfort negotiating.

6.8 Table VIII: Gender Differences in Job Mobility; Figure VI: Gender Pay Gap and Job Mobility

Read: Table VIII shows that gender differences in district-to-district mobility are small and do not rise after CBA expiration. Figure VI shows the pay gap persists among nonmovers and varies with local job-market outside options.

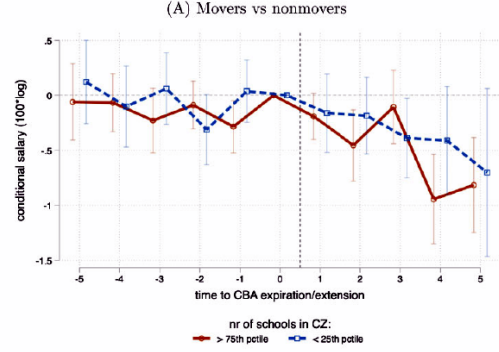
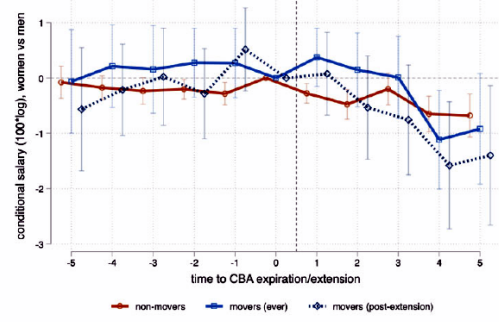
Economic message: Mobility is not the main explanation. Bargaining power may matter, but the wage gap appears even among teachers who do not move.

TABLE VIII
GENDER DIFFERENCES IN JOB MOBILITY

	All moves			Within CZ	Across CZ
	(1)	(2)	(3)	(4)	(5)
Female	-0.0014** (0.0006)	-0.0012** (0.0005)	-0.0009 (0.0007)	0.0004 (0.0003)	-0.0013** (0.0005)
Post Extension	0.0179*** (0.0012)	0.0036 (0.0023)	0.0007 (0.0028)	0.0024* (0.0014)	-0.0012 (0.0018)
Female $\times$ Post Extension	-0.0004 (0.0011)	-0.0004 (0.0012)	-0.0010 (0.0011)	-0.0006 (0.0008)	-0.0007 (0.0008)
District FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Exp., edu FE	No	No	Yes	Yes	Yes
N	413,920	413,920	413,801	418,383	412,931
# districts	247	247	247	247	247
Mean of dep. var.	0.0236	0.0236	0.0236	0.0102	0.0112

Notes. The dependent variable is an indicator for a teacher changing district (columns (1)–(3)), changing district within the same CZ (column (4)), and changing district and CZ in a given year (column (5)). The variable *Female* equals 1 for female teachers and the variable *Post Extension* equals 1 for years following the expiration of a CBA or its extension. Columns (2)–(5) include district and year fixed effects; columns (3)–(5) also include fixed effects for years of experience and for the highest education degree. Standard errors in parentheses are clustered at the district level. * $p \leq .1$, ** $p \leq .05$, *** $p \leq .01$.

Table VIII: Job mobility. Women do not become differentially less mobile after flexible pay begins.



(B) Gender pay gap and outside options

FIGURE VI
Gender Pay Gap and Job Mobility

Figure VI: Mobility. The gender gap persists among nonmovers and is linked to local outside options.

7 Short summary

- Wisconsin Act 10 weakened collective bargaining and introduced flexible pay once district CBAs expired.
- Before Act 10, salary schedules largely tied pay to seniority and credentials.
- Flexible pay created a female salary penalty among comparable teachers.
- The gap grows several years after CBA expiration.
- The gap is larger for younger/less senior teachers and under male leaders.
- Survey evidence shows women negotiate salary less often and report more discomfort negotiating.
- Teacher value added and observed mobility do not explain the gap.
- The paper provides field evidence that wage bargaining can widen gender pay gaps.

8 Conclusion

8.1 Core conclusion

Biasi and Sarsons show that flexible pay can widen gender wage gaps. When Wisconsin districts moved from rigid salary schedules to flexible compensation, women earned less than comparable men with similar experience, credentials, and assignments.

8.2 Economic intuition

Rigid salary schedules compress pay differences by tying wages to observable rules. Flexible pay allows individual negotiations and managerial discretion. If men bargain more often or bargain more successfully, and if counterpart identity matters, flexibility can translate into gender-specific wage growth.

8.3 Incremental contribution

The paper’s contribution is to provide field evidence on bargaining and gender gaps using an institutional shock that changes the pay-setting regime for the same occupation. It shows not only that flexible pay increases the gap but also where and why: in less schedule-bound districts, under male leaders, among younger teachers, and among workers who report lower bargaining comfort.

8.4 Policy implications

The results suggest that collective bargaining and salary schedules may narrow gender pay gaps by limiting the role of individual negotiation. If institutions move toward flexible pay, complementary policies - transparency, standardized negotiation protocols, manager training, or structured salary bands - may be necessary to prevent gender gaps from widening.

8.5 Limitations

The survey cannot be linked directly to individual wage records, so the exact share of the wage gap explained by bargaining behavior cannot be estimated. Leadership gender is not randomly assigned. The setting is public school teachers in Wisconsin, so magnitudes may differ in occupations with different bargaining norms, outside options, and evaluation systems.

8.6 Takeaway

The paper’s central lesson is that pay flexibility is not distributionally neutral. Removing rigid schedules can reward productivity and bargaining, but it can also create gender gaps when workers differ in negotiation behavior or when bargaining counterparts respond differently by gender.